

HAYLIE PEDERSEN

315-571-6223 | Boston, MA | pedersen.h@northeastern.edu
github.com/hayliepedersen | linkedin.com/in/haylie-pedersen | hayliepedersen.com
Availability: June 2027

EDUCATION

Northeastern University | Khoury College of Computer Sciences | Boston, MA Expected: May 2027
Candidate for Bachelor of Science in Computer Science GPA: 3.85/4.00 (Dean's List)
Concentration in Artificial Intelligence | Minor in Psychology
Highlighted Courses: Object-Oriented Design, Algorithms & Data Structures, Software Engineering, Systems, Databases
Activities: Northeastern Electric Racing, NU Women in Technology, Lead360 Program, Cheese Club

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C/C++, HTML, CSS, SQL
Frameworks/Libraries: React, Django, FastAPI, Flask, Express, Tailwind, Prisma, Pandas, JUnit, Streamlit
Software/Tools: AWS (Lambda, ECS, S3, RDS, ElastiCache, CloudWatch, Secrets Manager), Terraform, Docker, Redis, PostgreSQL, MySQL, Git, Figma

EXPERIENCE

Klaviyo | Boston, MA Jul 2026 – Present
Software Security Engineer Co-op

- Built the data collection layer for a risk-based authentication system, implementing a multi-region Redis cache end to end (client, hash interface, service functions) over AWS ElastiCache infrastructure provisioned in Terraform.
- Established telemetry across CloudWatch and Chronosphere to monitor service health and reduce incident triage time.
- Automated a reporting process with Python/Django tooling, eliminating manual entry across 18 required fields.
- Developed AI agent workflows for spec-driven development and code design review, alongside scheduled tasks that synthesize planner, git, and Slack activity into daily standups and weekly summaries.

Verizon Communications (NExT) | Boston, MA Jul 2025 – Dec 2025
Software Engineer Co-op

- Architected a full-stack telecommunications sentiment analysis platform using React/TypeScript and FastAPI/Python, deployed on AWS infrastructure (Lambda, S3, ECS, RDS PostgreSQL) with responsive design patterns.
- Maintained automated ETL workflows and data ingestion pipelines with daily triggers to aggregate and transform sentiment and network data from 5+ external sources, revealing geospatial hotspots across all 50 US states.
- Developed Isolation Forest anomaly detection system, flagging emerging issues for proactive trend monitoring.
- Enhanced performance and upgraded key packages, reducing load times by 30% with TanStack Query.

Northeastern Electric Racing | Boston, MA Sep 2024 – Present
Software Developer

- Collaborate with 60+ developers in an Agile environment to build a management platform serving 400+ members.
- Implement database schemas and API endpoints for operations such as role-based access and project tracking.
- Design UI components that adapt seamlessly across devices, enhancing accessibility and member engagement.

PROJECTS

ResearchCompass | React, TypeScript, Tailwind, Python, Flask Jul 2025

- Built a recommendation engine leveraging vector embeddings to match research abstracts with publication venues.
- Designed wireframes in Figma and built a responsive, user-friendly interface with React and Tailwind CSS.
- Integrated academic databases (DOAJ, arXiv) to provide targeted journal recommendations ranked by relevance.

PetFetch | MySQL, Docker, Pandas, Python, Flask, Streamlit Aug 2024

- Designed and implemented REST APIs in Flask, developing routes to support 15 CRUD operations.
- Containerized with Docker Compose, enabling local deployment and consistent environments across a 5-person team.

SKILLS & INTERESTS

Skills: Problem Solving | Motivation | Creativity | Adaptability | Group Leadership & Collaboration
Interests: Psychology | Hand Lettering | Traveling | Tea Tasting | Fitness | Fantasy Fiction